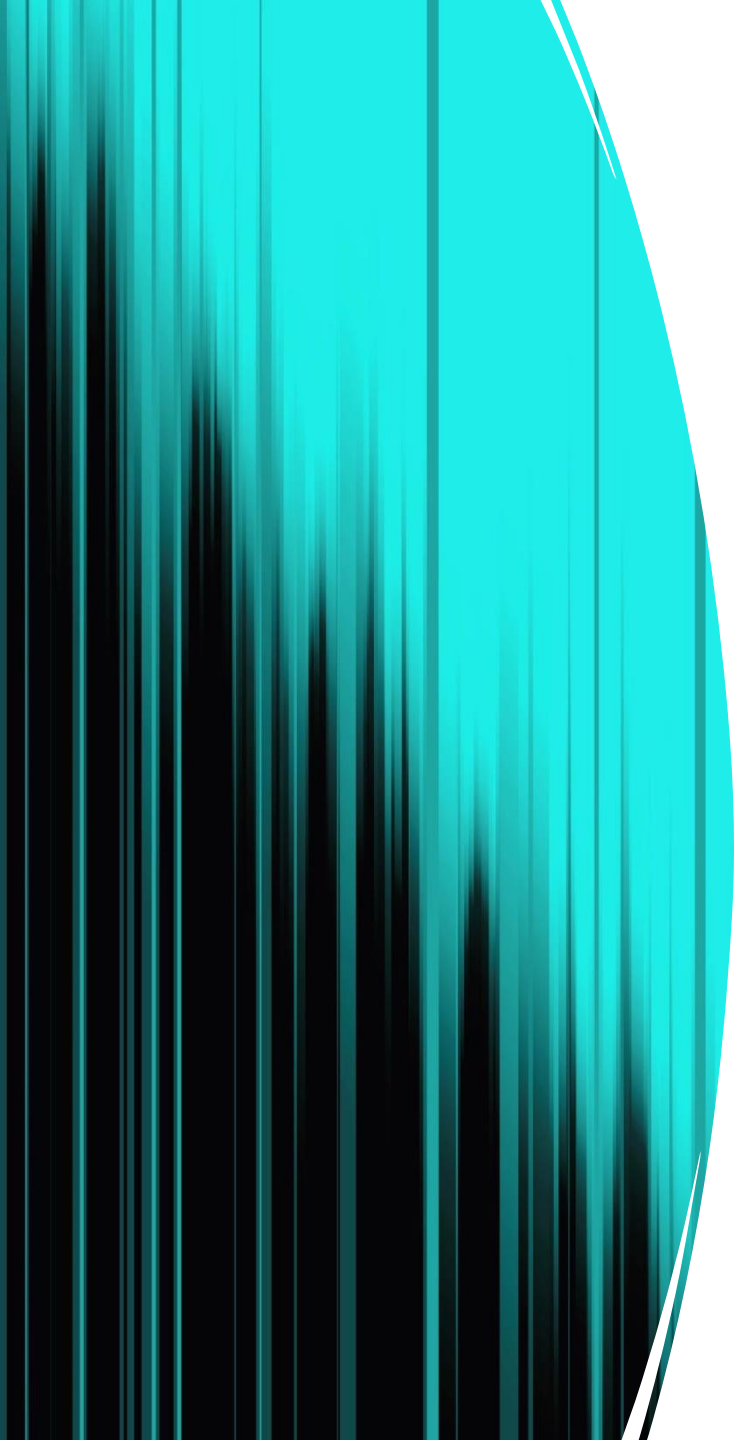




Welcome

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BCS-xxx

# Information Security

Dr. Irfan Uddin

# Dr. M. Irfan Uddin

- Associate Professor, DCS, AWKUM
- Over 20 years of teaching and research experience
- Published 130+ research articles (journals/conferences)
- Published 2 Patents (US)
- Published 3 Books
- Supervised: 80+ BS, 20+ MS, 5+ PhDs
- Interest: Deep Learning, Gen AI, Agentic AI, GPUs
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- WhatsApp: +92 3355764514

# Assessments

| S# | Name                                | Marks |
|----|-------------------------------------|-------|
| 1  | Assignments/ Quizzes/ Presentations | 20    |
| 3  | Mid Term                            | 30    |
| 4  | Final                               | 50    |
|    | Total                               | 100   |

# Contents of the course

- Motivation and overview of computer security
  - Security definition, policy, mechanisms, services and models
- Introduction to cryptography, symmetric/asymmetric encryption
  - DES/AES, RSA, MD5, SHA2
- Software security
  - Buffer overflow, heap overflow and string format bugs
- Operating system security techniques
  - Dealing with bad (legacy) codes: sandboxing
  - Multi-level security, file system security
- Internet vulnerability
  - Denial-of-service attacks, viruses, worms, Trojan horses
- Securing the Internet-Intrusion detection systems (IDSs): host- vs. network-based, signature vs. statistical detection, Firewalls, VPN and IPsec

# Books

- Kaufman, C., Perlman, R., & Speciner, M. (2022). *Network security: Private communication in a public world* (3rd ed.). Addison-Wesley Professional.  
<https://doi.org/10.5555/3152382>
- Stallings, W. (2017). *Cryptography and network security: Principles and practice* (7th ed.). Pearson Education. ISBN 9781292158587

# Other Referenced Books

- [Writing Secure Code](#), Michael Howard and David LeBlanc, Microsoft Press, 2002.
- [Security in Computing](#), Charles Pfleeger, Shari Lawrence Pfleeger, 3rd Edition, Prentice Hall, 2002.
- [Firewalls and Internet Security: Repelling the Wily Hacker](#), 2<sup>nd</sup> edition, by [William R. Cheswick](#), [Steven M. Bellovin](#), and [Aviel D. Rubin](#), Addison Wesley, 2003

# Acknowledgement

- Slides prepared from materials at:
- **EECS 350 Introduction to Computer Security**

# Contents

- Contents are available online at

<https://github.com/mirfanud/Fall2025>

- Or

<https://mirfanud.wordpress.com/courses/>



Lets start .....

